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LONG PAPER

# Designing spaces for every listener

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**Abstract** Inclusive design aims at objects and environments that are accessible, usable and comfortable for all people throughout their entire lifespan. In architecture, this aim is usually associated with physical accessibility. Yet acoustic qualities may considerably impact usability and comfort as well, especially in spaces for listening. This case study explores the notion of acoustic comfort for all in the context of university education. One auditorium, situated in a historic building and subject to renovation in the near future, is studied in detail: acoustic obstacles are identified in collaboration with user/experts and are measured in situ; specific interventions are proposed in consultation with building professionals, technicians and conservation specialists and are tested using dedicated acoustic simulation software. The study draws attention to the importance of acoustic comfort for all and offers a first view of which solutions are possible and desirable and how these can be obtained.

**Keywords** Architecture · Acoustic comfort · Inclusive design

## 1 Introduction

The past decade has witnessed a major shift in how human limitations are interpreted [1]. Whereas disability used to be considered as a consequence of physical and/or cognitive limitations, it is increasingly recognised that the (built) environment holds responsibility for the exclusion of people as well, in other words that the environment can create handicap situations. This view is complemented by a shift in professional attitudes: design for special needs, which focuses on a specific target group, gradually makes way for universal design, design for all or inclusive design. Inclusive design aims at objects and environments that are accessible, usable and comfortable for the full range of people, independently of their capacities and limitations, throughout their lifespan.

While all people differ in terms of capacities and limitations, human functioning can be generally described by the ergonomic principle [2]. This principle assumes that people perceive something in their environment, interpret what has been perceived, act on the basis of this interpretation and remain healthy while performing these three 'ergonomic tasks'. Functioning in the built environment involves at least three modes of perception: seeing, touching, and—the focus of this paper—hearing. Hearing provides auditory information about the environment, which is vital for human interaction and knowledge transfer, and essential for spatial orientation [3]. Good conditions for auditory communication require attention to the acoustic qualities of a space and its sound amplification system. It is especially important for people with a hearing or visual impairment, but also for the growing group of elderly people or when an extra language effort is needed.

So far, however, studies on inclusive design in architecture have paid little attention to hearing. They tend to focus on physical accessibility of buildings and spaces, but rarely address their acoustic qualities. The same holds for accessibility regulations. Some prescribe or recommend the instalment of an induction loop, but since this serves only

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those users who wear a hearing aid, it does not guarantee access, usability, and comfort for all. More generally, guidelines exist for background noise level and reverberation time—two factors that considerably impact the acoustics in a space—yet their presence in codes and norms (and in building practice) varies considerably. In some countries, the guidelines are imperative, while others consider them as recommendations [4]. Across the board, however, little attention is paid to the diversity of people's hearing capacities and needs.

The scarcity of attention to acoustic comfort in designing inclusive environments is highly regrettable, because users with deficiencies in hearing ability could be subjected to extra difficulties in verbal communication [5]. The study reported in this paper attempts to contribute to the knowledge base by exploring the notion of acoustic comfort for all. The term *acoustic comfort* is used here to denote the state of satisfaction with the acoustics of the surrounding environment. Just like thermal comfort, it differs from one space to another and depends on the activity taking place in that space. For instance, the preferred air temperature in a day care centre differs from that in a beer factory. Similarly, for some places, a certain degree of speech privacy may be expected (e.g., open plan offices), whereas for others good speech intelligibility is essential for their use (e.g., auditoria). Furthermore, music typically needs more reverberation than speech. In this paper, the notion of acoustic comfort for all is explored in the context of university education, a context where hearing well is vital to the core activity. After a brief overview of related work (Sect. 2), the paper introduces the aims and methodology of the study (Sect. 3) and presents the major findings (Sects. 4, 5, 6) followed by a summary and lessons learned (Sect. 7).

## 2 Related work

Renovating historic buildings or protected monuments from the perspective of inclusive design is not a completely new topic of interest [6]. Often, however, these renovations are limited to improving the building's physical accessibility, while leaving its acoustic conditions unchanged.

In some cases, this might be due to the fact that conservation of the protected monument also means conservation of its acoustic conditions, out of respect for or fear to change 'the excellent acoustics'. This applies especially to places for musical performance: historic concert halls or theatres with respected acoustic qualities such as the Paleis voor Schone Kunsten (Centre for Fine Arts) in Brussels [7] or UNESCO protected Baroque theatres like in Český Krumlov [8].

There are a lot of examples, however, where renovation of the acoustic conditions should be considered in view of new functionality or new demands. A case in point is the refurbishment of churches. When the original architecture is restored without special acoustic measures, the historic acoustic conditions of long reverberation and poor speech intelligibility are restored as well (see also Sect. 5). This is exactly what happened to the Sint-Jan-de-Doper church in Leuven [9]. After restoration, the acoustics turned out to be by far worse than before: they became problematic as they still are today. During the restoration, no care was taken of the changed functionality: unlike in the seventeenth century, nowadays speech intelligibility is important. If such a church—or other historic building—is refurbished so as to serve for music performance purposes, people with hearing limitations should be assisted by the changes, not hindered.

## 3 Aim and methodology

### 3.1 The Grote Aula

The case study reported in this paper was conducted within the context of the Katholieke Universiteit Leuven (K.U.Leuven) [10]. Within this university, one auditorium was selected to explore the notion of acoustic comfort for all. The Grote Aula (large auditorium) was chosen because of its frequent use for lectures and musical events, the complaints of user/experts about its poor acoustics, and the plans to renovate it in the near future.

The Grote Aula of the Maria-Theresia College is situated in the old city centre of Leuven. In 1778, Maria-Theresia accommodated the seminar of theology in this former Jesuit College. The right wing in the inner courtyard, which still accommodates the Faculty of Theology, was added in 1786–1787. The Grote Aula (Fig. 1) and Kleine Aula (small auditorium) followed in 1825–1827. These neo-classicist auditoria, designed by Martin Hensmans, have a half-round ground plan and are on one floor. Since 1975, the Grote Aula is officially listed as a protected monument, together with the entire Maria-Theresia College and the neighbouring Pauscollege (Pope's College).

At the time of writing, the 495 seats of the Grote Aula accommodate undergraduate and graduate students on a daily basis. Once a week, students of the University for the Elderly attend lectures here, and at night, or between semesters, the auditorium is available for music activities. Both types of events—lectures and music events—require different acoustic conditions: a speaker demands a much shorter reverberation time than a music ensemble. User/experts especially complain about the difficulty of clearly understanding the speaker during lectures. Comparison with other auditoria at the K.U.Leuven confirmed that the acoustic conditions in the Grote Aula show considerable room for improvement. The planned renovation in 2009

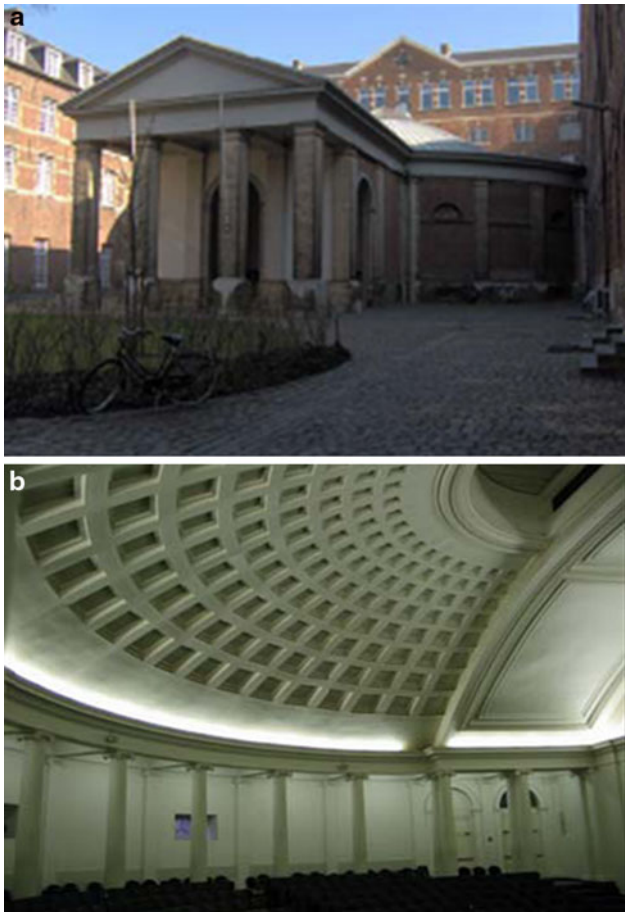


**Fig. 1** **a**, **b** The Grote Aula: exterior (photograph: K. De Leye & E. Dooms) and interior (photograph: M. Rychtáriková)

provided extra reasons to submit this auditorium to an in-depth analysis and to propose interventions that improve acoustic comfort for all.

### 3.2 Methodology

Inclusive design confronts architects with a new challenge. When designing environments that are targeted to be accessible, usable, and comfortable for more people, a human-centred process should replace the design-centred process that characterises non-inclusive design [11]. Thus, towards providing acoustic comfort for all, primary attention needs to be given to the users and their diverse hearing capacities and needs.

To start with, a series of in-depth interviews were conducted with various user/experts—students and personnel with a hearing impairment, students with a visual impairment, and students who attend the University for the Elderly. These interviews provided a rough idea of the acoustic comfort in the K.U.Leuven auditoria, which motivated the selection of the Grote Aula for the case study. At the same time, they allowed to start mapping the hearing needs and preferences of people with diverse capacities and limitations.

Subsequently, an attempt was made to address these needs and preferences in the context of the Grote Aula. The auditorium's present acoustic conditions were measured and evaluated. In order to improve these conditions, architectural interventions that respect the building's historic character were searched for and simulated. In addition, assistive technologies that could further improve the acoustic comfort in the auditorium were inventoried, so as to include people with and without a hearing impairment.

The following section will outline the findings of this study without going into too much technical detail.

## 4 User/experts speaking

In order to get a view of diverse hearing needs and preferences in the context of university education, six students with a hearing or visual impairment were interviewed. These user/experts in the domain of acoustic comfort were contacted via the Working Group Students with Disabilities of the K.U.Leuven and participated voluntarily in the study. In addition, a focus group interview was conducted with representatives of the University for the Elderly. All interviewees were asked to nominate the best and worst auditorium in terms of acoustic comfort and to indicate problems they experience when attending lectures. The interviews were audio-taped, the tapes were transcribed, and the transcriptions were summarised.

While opinions on the best auditorium varied, several user/experts labelled the Grote Aula as the worst. The 'strong resonance' or 'long reverberation' especially is considered as negatively impacting the acoustic comfort of this auditorium.

Across the board, 'resonance' and 'reverberation' are cited most frequently by all interviewees as causing hearing problems during lectures. Students with a hearing impairment need a clear and high-quality signal in order to be able to understand the speaker, because their remaining sense of hearing is limited. Students with a hearing aid often use an FM listening system during lectures. When the speaker speaks into the microphone, the speech signal, which includes apparently too much of the reverberant sound, is sent to the hearing aid. According to one student, it is so difficult to understand speech in the Grote Aula that all her effort has to go into comprehending, leaving no room to write her own notes. In other words, the FM system does not offer a solution; better acoustic conditions would help to make the system work more properly and less subjected to sub-optimal conditions, such as a wrong distance or direction from speaker to microphone.

Also, students with a visual impairment complain about auditoria with long reverberation. In these spaces, they receive so many inputs that orientation becomes extremely difficult. Since they move relying on their sense of hearing (and touch), it becomes almost impossible for them to orient themselves. Moreover, students with a visual impairment tend to audio-record lectures for later use; yet, in these auditoria, this is of little use since the sound quality of the recordings is very low.

Besides reverberation, the user/experts mention several other factors that affect acoustic comfort. Background noise—caused by chatting students, for instance, or cracking wooden seats—is very disturbing, especially for students who use hearing aids: these devices amplify the sound without any selectivity. Other elements relate to the speaker: articulation quality, familiarity with the available technology, and awareness to repeat questions asked in the audience.

## 5 In situ measurements

As the Grote Aula was identified by several user/experts as the worst auditorium in terms of acoustic comfort, a detailed view of its present acoustic conditions was necessary. To this end, measurements were conducted in situ, which enabled to derive several acoustic parameters. Based on the results of these measurements, the auditorium was assessed in more detail and interventions for improvement were proposed. The following paragraphs briefly introduce the reverberation time and the speech intelligibility—the most important acoustic parameters in the context of this paper—without the technical details, and summarise the major results of the measurements and their implications.

### 5.1 Reverberation time

As mentioned above, long reverberation strongly affects acoustic comfort in a negative way. The acoustic parameter that models this phenomenon is the reverberation time (RT). A long RT (of several seconds) extends syllables so that they overlap, and thus negatively impacts speech intelligibility [12]. This is the case in spaces with hard, reflecting walls and ceilings. A long RT causes the background noise level to rise, which further reduces speech intelligibility.

A too long RT increases the background noise and makes listening very tiring and perhaps even impossible. For people with a hearing impairment, the maximum allowable RT is significantly shorter. For people with a visual impairment, a shorter RT is crucial too. In a room with a too long RT, these people have difficulty locating from which direction the sound is coming. However, the RT has also a lower limit. Too much absorption in a space, especially in auditoria, weakens the early sound reflections such that the sound level decreases too strongly in the back of the room [12].

According to the performed measurements, the RT in the Grote Aula is 2.40 s, which was to be expected given its size (3,588 $m^3$) and materials (hard surfaces and leather seats). This is too long—for lectures and for music—and should diminish drastically in order to provide acoustic comfort for all. In a space of this size, 1 s would be ideal for speech, while music would demand some 1.8 s. These values are based on acoustic practice [13]. In fact, no strict 'standards' exist; a certain degree of freedom can be left since good room acoustics involves more than a discussion only on RT.

### 5.2 Speech intelligibility

Speech intelligibility is an intrinsic characteristic of speech transmission for a certain position of source and receiver. The size and shape of the space, and its finishing, furnishing, and occupation, clearly affect the strength and the quality of the transmission of the spoken signal. Speech intelligibility will also depend on the specific location of both speaker and listener, on the strength and quality of the spoken signal, and on its level relative to the background noise level [14].

The speech transition index (STI) [15] of a space varies between 0 and 1; the higher the value, the better the speech intelligibility. An STI of 0.6 is considered as desirable, yet complete intelligibility of sentences requires 0.75. Research on speech intelligibility with elderly (suffering from a limited hearing loss) and with non-native listeners and speakers, shows that an increase in STI of minimum 0.10 is desirable, yet in many cases 0.15 or 0.20 may be necessary [16, 17].

From the measurements, the STI in the Grote Aula varies between 0.36 and 0.50. Across all positions, the average is 0.44, which is usually experienced as poor. The value diminishes in the back, since the sound level reduces with the distance from the source; the more absorption in the space, the larger this reduction. Moreover, the farther from the speaker, the more reflections (with walls, ceiling) have taken place. If strong reflections arrive at the listeners' position with a considerable delay relative to the arrival of the direct sound (more than 50 ms), they are detrimental to the speech intelligibility. Judging from listening to and verifying the recorded responses or echograms (Fig. 2), this did not happen to be the case.

When using the current audio system in the Grote Aula, the measured STI varies between 0.37 and 0.69, with an average of 0.48 (before: 0,44). This suggests that the system does not perform properly. A better installation that

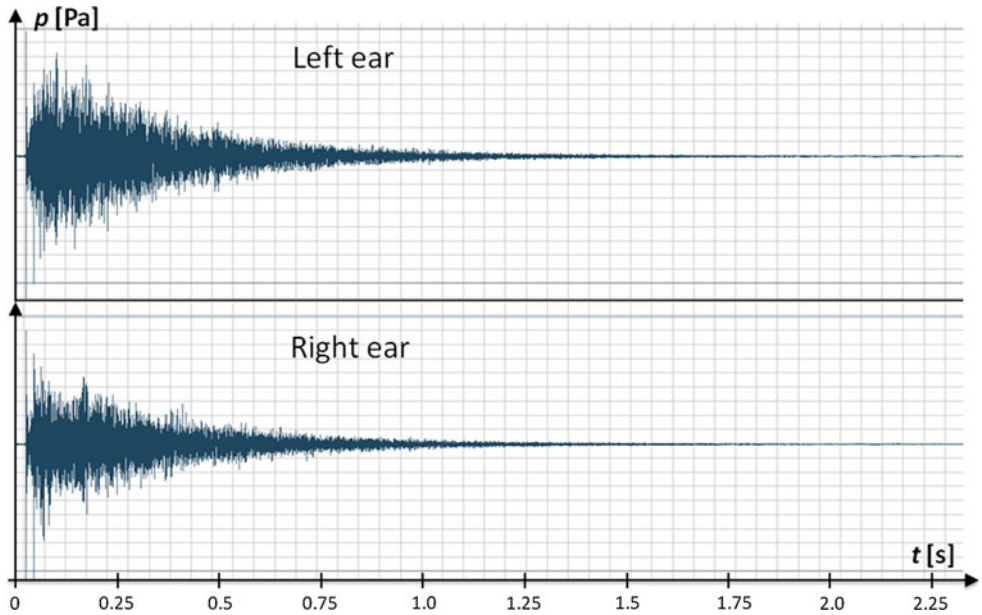


**Fig. 2** Example of a binaural echogram measured at the listener's ears at a listening distance of 15 m from the sound source (recorded using the artificial Head Acoustics head and torso system). This echogram displays the sound pressure's time history, registered at both ears after an impulsive sound excitation near a lecturer's position. No particular individual strong reflections show up after a 50 ms time delay from the arrival time of the direct wave front (50 ms is regarded as the minimum delay needed to perceive strong isolated reflections as echoes)

distributes amplified sound in a proper way across the entire auditorium should considerably improve the speech intelligibility. The improvement of the averaged STI is only 0.04 with the actual system, which is far from optimal. A good system should deliver at least a 0.15 increase. The weak performance is due to poor audio system components, bad location, lacking directivity, and non-optimal pointing of some of the loudspeakers.

## 6 Improvements

### 6.1 Spatial interventions

Judging from the measurements in situ, providing acoustic comfort for all in the Grote Aula will require major improvements. A first set of improvements comes down to adding absorption, so as to reduce reverberation and increase the speech intelligibility.

As the auditorium is a protected monument, architecturally modest solutions are needed to provide extra absorption. In the case of a new building, many more possibilities exist to create a favourable acoustic climate.

Given the double function of the Grote Aula—lectures and music—the possibility was explored of applying movable interventions, which can be adapted to the activity taking place. The following sub-sections describe the interventions which were considered.

#### *6.1.1 Replacing the leather-covered seats by lightly upholstered seats with cloth covering*

Lightly upholstered seats covered with air-transparent cloth provide more absorption than the present leather-covered seats. The absorption increases with the thickness and can even reach the absorption of a seated person, thus weakening the influence of the level of occupation of the auditorium. Such seats are typically used in movie theatres or concert halls, but are not of application in a student environment.

#### *6.1.2 Replacing the linoleum floor by carpet*

The Technical services of the K.U.Leuven consider replacing the present linoleum floor by carpet. Carpet has considerable advantages in terms of acoustic qualities, but also considerable disadvantages in other respects. Advantages of carpet include higher absorption values than most other floor materials and reduction in impact sounds of walking persons; this results in less disturbance when someone enters the auditorium after the lecture or concert has started. However, the aim of the current study being to provide comfort for all, the use of carpet raises several questions. Carpet is not only difficult to maintain, it also makes the auditorium less accessible for wheelchair users and less comfortable for persons with certain allergies.

#### *6.1.3 Applying an acoustic plaster to the curved wall in the back*

Reflections on the walls—and certainly on the curved wall in the back of this auditorium—sustain late reflections and increase the RT. In order to avoid this effect, it is advisable to apply extra absorption to these surfaces. In order to respect the original interior, the absorbing layer should be as thin as possible. Since the present finishing in the Grote

Aula consists of plaster, one could opt for applying an acoustic plaster on an acoustic absorption layer along the curved wall in the back. Systems exist in which the total thickness is limited to 40 mm. They provide relatively high absorption, except in the lower frequencies. Note that the absorbing plaster should not be painted with an airtight covering paint film, as this would reduce the access of the sound waves to the porosity of the material and thus the absorptive efficiency.

#### *6.1.4 Placing removable sound-absorbing wall panels covered with cloth*

Another visually acceptable solution to apply absorption to the curved wall is by placing removable wall panels. The core of these panels is made of sound-absorbing foam and is surrounded by a steel frame for solidity. The whole is covered by open-cell textile that extends around the frame such that it remains invisible. Because the panels are easily installed but also removed, the wall is affected as little as possible. They can be hung onto rails or magnetically fixed onto a metal case or partition. Overall, the thickness of this solution amounts to 60–50-mm panel plus 10-mm cavity. The absorption is slightly better than that of an acoustic plaster and would offer a considerable improvement compared to the present wall finishing.

#### *6.1.5 Hanging curtains along the wall*

An even more flexible solution is hanging curtains along the curved wall. The main advantage of curtains is that they can be closed when absorption is needed and opened when, for instance, a music ensemble wishes a longer reverberation time. The absorption that curtains provide depends on the distance from the wall—the larger the distance, the higher the low frequency absorption—and on whether they are hanging flat or folded, the latter being the most effective on sound absorption.

#### *6.1.6 Applying a thin acoustic plaster to the cupola and vault above the podium*

Reflections via the cupola and barrel vault (above the podium) also cause strong late reflections, but due to a strong depth profile, these are highly scattered. However, the plastered ribs of the ceiling refrain from application of thick sprayed porous plasters, as this would destroy the texture of the ceiling surface. Only a reduced thickness (10–20 mm) would be acceptable, which would still mean acceptable high frequency performance (speech) but a loss of low frequency performance (sub-optimal for music).

#### *6.1.7 Applying an acoustic plaster to the ceiling above the gallery*

Also, the ceiling above the gallery may be treated with an acoustic plaster to increase the absorption in the auditorium. Since there is no special texture here, a thicker plaster can be applied. For instance, one could opt for the acoustic plaster on an acoustic absorption layer, as suggested for the curved wall, which provides significantly more absorption than an acoustic plaster as such.

### 6.2 Simulation and analysis

In order to test the impact of the proposed interventions on the spatial acoustics of the auditorium, a 3D model of the Grote Aula was developed in AutoCAD, and subsequently exported to an acoustic simulation software (Odeon). This enabled to calculate various acoustic parameters, the results of which are discussed and illustrated below.

The acoustics in the auditorium were simulated in different steps, starting with the calibration of the model to the current situation. When testing the 3D model without any improvement measures, the results obtained were 2.48 s for the RT and on average 0.45 for the STI, which correspond to the measurements in situ. As Fig. 3a illustrates, the middle rows enjoy a slightly better STI than the rest of the auditorium. For the front rows, this may be explained by the fact that the difference in path length between the direct sound and the reflections is larger, which results in later sound reflections that diminish the STI. On the other hand, the sound level decreases with the distance from the speaker, which explains the lower STI in the back row that is only served by a multitude of weaker late reflections.

After calibrating the model to the current situation, tests were conducted step-by-step on (combinations of) the interventions proposed earlier, placing microphones and sources in the same position as for the in situ measurements. Table 1 summarises the RT and STI for all tests. By way of example, Fig. 3b–d displays the STI for three combinations of interventions. For instance, when combining cloth-covered seats with a carpet floor (Fig. 3b), the average STI amounts to 0.51, which is a considerable improvement compared to the current situation. However, the minimum requirement for good speech intelligibility is still not attained. Moreover, one should not forget that this improvement comes at the cost of hygienic and health issues related to the carpet.

In order to get an overview of these different aspects and issues, Table 1 schematically maps the performance on different criteria for the different interventions:

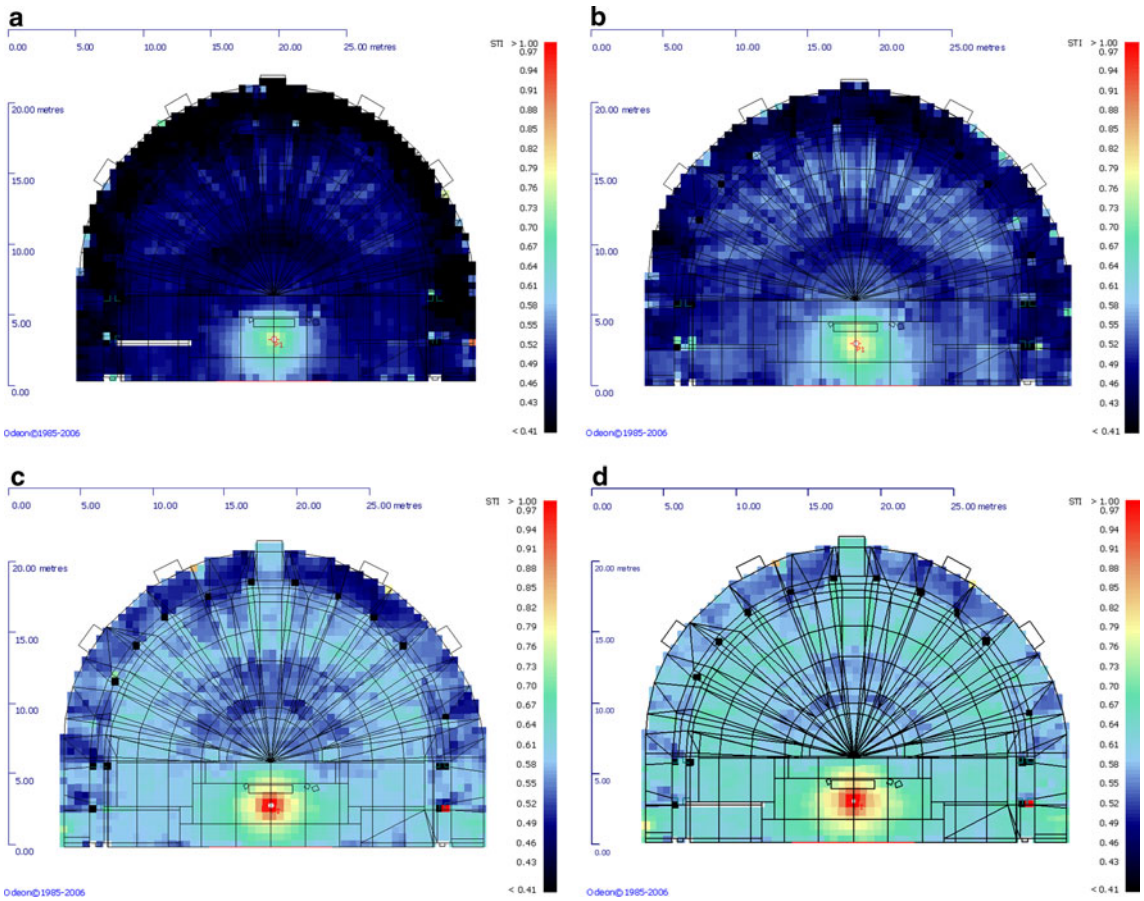


**Fig. 3** **a** STI current situation; average: 0.45. **b** STI seats + floor; average: 0.51. **c** STI seats + floor + wall (plaster); average: 0.58. **d** STI seats + floor + wall (plaster) + cupola and vault; average: 0.62

**Table 1** Performance of interventions

| | Current | Interventions | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | **1** | 1,**2** | 1,2,**3** | 1,2,**4** | 1,2,**5** | 1,2,3,**7** | 1,2,3,6,**7** | 1,**3** | 1,**7** |
| **RT** | 2.48 | 2.44 | 1.72 | 1.18 | 1.10 | 1.26 | 1.17 | 1.01 | 1.90 | 2.22 |
| **STI** | 0.45 | 0.46 | 0.51 | 0.58 | 0.58 | 0.59 | 0.59 | 0.62 | 0.52 | 0.48 |
| **Speech** | | | | | | | | | | |
| **Music** | | | | | | | | | | |
| **Maintenance** | | | | | | | | | | |
| **Flexibility** | | | | | | | | | | |
| **Replacement** | | | | | | | | | | |

*Key*: *black* weak, *grey* moderate, *white* optimal

- speech: RT of ca. 1 s and speech intelligibility > 0.6;
- music: RT of ca. 1.8 s;
- maintenance: how easily materials can be cleaned and how fast they get dirty;
- flexibility: adaptability to change in use (music/speech);
- replacement: how easily solutions can be removed or replaced.

Note that the score for the latter three applies to the intervention in bold only.

The RT clearly decreases as absorption increases. Solutions that are labelled as moderate for speech do not meet the requirement for RT and STI, but may come close. Solutions marked as weak are to be avoided, as the RT is way too long and the STI too low. Solutions that do not meet the requirements for music are marked as moderate,

because the echo in the space is either too strong or insufficient for music to sound optimally.

Cloth materials or covers are moderate solutions with regard to maintenance; they easily absorb dirt that is difficult to remove. Carpet is even worse, as it quickly becomes dirty in use. Moreover, it is less comfortable for wheelchair users or people with allergy.

Curtains are considered flexible because they can easily be opened or closed depending on the activity taking place (lecture/music). Wall panels can be removed, but this is much more laborious. Plasterwork cannot be removed or replaced without damaging the underlying structure and is therefore marked as weak with respect to flexibility and replacement. Curtains and wall panels are easily removable, while the moderate solutions require more effort.

Depending on the priorities set, one can select the most appropriate solution from the scheme above. Obviously, the fact that the Grote Aula is protected as an historic monument does play a role in this selection. However, whatever solution is selected, increasing absorption alone will not automatically lead to acoustic comfort for all. As the table indicates, none of the solutions perform optimally for reverberation and speech intelligibility—even less so when taking into account the more stringent acoustic conditions imposed by people with a hearing or visual impairment. Since these are not attainable in the context of the Grote Aula, the acoustic comfort needs to be further improved in a technological way.

## 6.3 Assistive technologies

By applying extra absorption, the sound level will decrease considerably. As a result, the sound signal will be too weak to be perceived by the audience in the back rows of a large auditorium such as the Grote Aula. In combination with background noise, distinguishing the voice of the speaker becomes highly difficult. Persons with a limited hearing capacity (e.g., elderly, persons with a hearing impairment) experience even more nuisances from background noise, because their hearing aid is unable to completely filter out the disturbing sound.

In order to address these problems, technical aids are needed. In the context of the conducted study, several assistive technologies were investigated that bring the sound signal to the listener in a more direct and/or pure way.

### *6.3.1 Induction loop*

In case of an induction loop, the sound is recorded through a microphone and sent to an amplifier. The latter sends an electrical current through the induction loop, i.e., an insulated copper cable, which in turn creates a fluctuating magnetic field. The magnetic signal can be received by the hearing aid when turned into the T-position. An existing sound installation, TV, or radio can directly be connected to an induction loop amplifier as well.

Because the signal is no longer perceived via sound waves travelling through the auditorium, but transmitted directly to the hearing aid, the influence of the space (e.g., reflecting surfaces) on the sound signal is largely eliminated. Obviously, high-quality signal capitation is necessary, but the distance to the speaker, background noise, and reverberation time will no longer raise problems for persons wearing a hearing aid.

An induction loop can be either small (individual loop) or large (loop around the auditorium). The former hangs around the listener's neck and is plugged into the exit of a headset. The latter can be integrated in the floor or ceiling. To guarantee an optimal signal, the width of the loop should be limited. In a larger space, it is recommended to use multiple compartments, so as to avoid large differences in the strength of the magnetic field. There is no signal right above or below the cable, so all seats should be located within the loop. Moreover, the cable should not be installed higher than 3 m above or lower than 0.5 below the floor, because the useful signal may become too weak. In buildings where, for instance, reinforced concrete is present, a stronger amplifier may be needed as the reinforcements are magnetically absorbing.

### *6.3.2 Headsets*

A solution that brings the sound directly and thus in a more pure form in the ear for persons without hearing aids is the use of headsets. In this case, the sound is transmitted through the electro-acoustic device (directive microphone + amplifier) directly to the headsets. As a result, the characteristics of the room have a minor impact on the quality of the sound signal and all listeners receive a signal of equal quality, whether seated in the front or in the back of the auditorium.

However, in some extreme cases, the recorded voice signal might still be influenced by strong sound reflections that may be amplified by the sound system and passed on to the listeners (effect of 'ringing'). This happens, for instance, when the interior surfaces are highly reflective and the microphone is less directive or badly positioned. Architectural measures and assistive technologies thus need to be considered at the same time.

Headsets can also be used for translation by an interpreter. Moreover, individual induction loops can be plugged into the exit, thus providing better intelligibility for persons with and without a hearing aid.

#### 6.3.3 Loudspeaker columns

The design of the public address system plays an important role. The specific loudspeaker quality, the number of loudspeakers, their directivity and positioning in the (mostly enclosed) space are crucial to reach the necessary speech intelligibility. A good—reasonably flat—frequency response is needed to obtain a natural sound experience, while adequate directivity is necessary for beaming the sound to the listening area.

In order to improve the directivity column, loudspeakers are used. These focus on the horizontal plane perpendicular to the column's axis. Yet, in order to direct the sound to the audience, they have to be mounted in an inclined way, which may not be the most esthetical solution. However, actual signal technology allows for implementing digital delay lines. Following the physical principle of Huyghens allows to redirect the beam downwards and to focus on the audience even if the loudspeaker is mounted vertically against a wall [18]. In this way, the visual disturbance by inclined speaker columns can be avoided, and the visual intrusion in historic environments is minimised.

In these Digital Directivity Control (DDC) systems, the characteristics of every individual loudspeaker in the column are software-driven, so as to give the sound bundles, the correct opening angle, direction, and focus distance. The vertical mounting has an extra acoustic advantage as well. In fact, part of the sound a loudspeaker column produces is directed backwards. Yet, in these DDC-loudspeakers, the backward-directed sound lobe is reflected in the same direction as the direct sound and thus reinforces it. In case of a traditional physically inclined loudspeaker, on the contrary, the backward-directed main sound lobe will cause late ceiling reflections which are mostly detrimental to speech intelligibility.

The current permanent loudspeaker installation in the Grote Aula consists of two standard non-directive loudspeaker boxes positioned left and right on the stage. By applying precisely directed DDC loudspeaker columns, the intelligibility could improve considerably without visual disturbance. Many examples of such applications in sensitive architectural and historic spaces can be found [19]. For sufficient speech intelligibility, an STI of at least 0.65 should be the purpose.

#### 6.3.4 Summing up

Several assistive technologies may improve speech intelligibility in an echoing space. Within the gamut of loudspeakers alone, multiple types and possibilities exist. This study has considered only interventions that can bring the sound signal to the listener with better quality.

An induction loop and the use of headsets avoid sound waves having to cover a path through free space before reaching the listener. As a result, the poor acoustic qualities of the space have less impact on the quality of the signal transmitted.

Loudspeaker installations can also solve this problem by directing the sound to avoid the reflecting surfaces and creating a well-distributed direct field. Applying extra absorption makes this easier, but a compromise is needed in relation to music activities.

Unlike the interventions to increase absorption, the proposed assistive technologies have not been tested with simulation software because of the laborious and very specific character of the subject matter. If specific performance requirements are formulated in advance, these calculations or simulations can be performed by or for the suppliers of these technologies.

## 7 Summary and lessons learned

In architecture, inclusive design aims at environments that are accessible, usable and, comfortable for all people. This aim is usually associated with physical accessibility, yet acoustic qualities may considerably impact usability and comfort as well. By way of a first step, this study explored the notion of acoustic comfort for all in university education, a context where hearing well is vital to the core activity.

The experiences of diverse user/experts revealed that the Grote Aula performs very poorly in terms of acoustic comfort. Measurements in situ confirmed the need for radical improvements. Several interventions regarding material use were proposed—taking into account the auditorium's historic character and dual use—and tested in terms of effectiveness via a simulation model. The performed analysis shows that the ideal solution does not exist; what is most appropriate depends on the priorities set (lectures versus music, maintenance, flexibility, replacement). Since none of these interventions guarantees acoustic comfort for all, including people with a hearing or visual impairment, additional assistive technologies are proposed: a better audio system, a headset connection and an induction loop. In addition, speakers' awareness should be raised regarding the importance of articulation, correct technology use, and repeating questions asked in the audience. In short, offering acoustic comfort for all in the Grote Aula requires a holistic interplay between spatial interventions, assistive technologies and human protocols.

Currently, the Technical services of the K.U.Leuven are negotiating with the Built Heritage authorities, but most likely will implement the following interventions: upholstered seats with cloth covering; carpet strips in the front,

the gallery and the corridors in between the seats; acoustic panels along the curved wall in the back; a better sound installation; and, most probably, an induction loop. The case study presented in this paper lays at the basis of all these decisions and played a crucial role to convince Built Heritage to step into the story (especially with regard the panels in the back).

The Grote Aula is a protected historic monument, and since every monument is protected in a specific way, general guidelines are difficult to derive. Nevertheless, the outcome of the case study largely transcends the planned renovation of this specific auditorium. The issues inventoried and addressed in this study may serve as points to consider when (re)designing other auditoria in the future. Moreover, the solutions explored in the context of the Grote Aula may be usable in other renovation projects. The same holds for the approach of looking for flexible changes by using acoustic predictions. Judging from the acquired experience, room-acoustic simulations provide a convenient method to evaluate particular alternatives in the context of historic buildings. As such, the study contributes to sensitisation and knowledge transfer, in that, it makes students and professionals in the building sector aware of the importance of acoustic comfort for all and offers a first view of which solutions are possible and desirable, and how these can be obtained.

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